SPE-229502-MS

Autonomy and Artificial Intelligence: Transforming the Role of Robotics and Drones in Subsea Operations

A. Anderson, Oceaneering International, Aberdeen, Scotland; I. Shuaib, Director, Arab Development, Abu Dhabi, UAE

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

Abstract

Artificial intelligence (AI) is changing the face of the offshore oil and gas industry. As companies apply AI and machine learning (ML) to subsea operations, it is increasingly apparent that technological advances can transform subsea operations in the Middle East and around the world.

Today, Uncrewed Intervention Drones (UIDs) and robotics are delivering efficiencies by:

- Improving inspections, saving time, and reducing costs by replacing vessel-based ROVs and Rules based traditional AUVs.
- Enhancing operational efficiency with expedited response times for inspection and maintenance
- Employing predictive analytics to improve acquisition speeds and drone endurance
- Delivering higher-quality data and more precise performance for fewer mission re-runs and faster decision-making.

Introduction

The offshore energy industry is integrating advanced UIDs to enhance operational efficiency, reduce costs, and improve decision-making. Soon, UID systems will be monitored remotely and controlled via onshore facilities to enable continuous inspections, infrastructure monitoring, and light intervention tasks to deliver be a highly efficient and scalable solution that enhances sustainability, safety, and cost-effectiveness.

The patented features detailed in this paper will be transformative for subsea operations. Using advanced data compression techniques to compress images up to 400x and reconstruct them using AI is game-changing because it enables high-quality image transmission and significantly reduces the bandwidth required. The use of advanced computer vision techniques to create less dense 3D point clouds that can be transmitted and reconstructed with up to 91% pixel-to-pixel accuracy further improves operational efficiency. This enhanced capability will be production-level coded for incorporation within systems in 2025/2026, effecting a significant leap forward in subsea robotics. This industry-first feature will be designed into new units and will allow existing systems to be easily updated.

Statement of Theory and Definitions

Implementing edge computing techniques on subsea robotics systems is the key to unlocking complexity in autonomous behaviors. Because edge computing places computation and data storage close to the source of data collection, it reduces latency and bandwidth use. This revolutionizes subsea robotics by enhancing real-time data processing onboard the subsea robot, removing spurious data, reducing latency, and enabling real-time onboard decision-making by applying predictive analytics to the data. With edge computing, subsea drones will have increased capacity to execute more complex tasks independently, and because edge computing enables localized sophisticated analytics, immediate insights can be used to inform the vehicle flight control system and improve the efficiency and accuracy of underwater surveys and inspections.

Effective subsea communication is essential because it is critical for:

- Human operators to exercise influence and control of underwater and surface systems
- Coordination and collaboration among multiple vehicles/robots
- Data exchange and conveying mission updates
- Mobility in challenging environments
- Integration with wider networks to allow gathered data to be shared.

The high attenuation of radio waves in seawater severely limits communication distances. Current primary underwater communication methods utilize acoustics or optical technology. Acoustics achieve a range up to 100 km with a very slow data rate (<5kbps), and optical technology achieves higher data rates (up to 100Mbps) but within a range limited to <100m.

To address this problem, Oceaneering has designed robotic systems that determine how to achieve communication by working within the confines of the systems available and contending with technology constraints in multiple ways that include:

- Resilience to loss of communication by being able to continue a mission effectively and safely
- Ability to communicate using multiple communication mediums
- Adeptness to perform behaviors that improve or benefit communication at the right time.
- Capacity to adjust the priority of messages being sent from the robot to scale down the bandwidth required to match the bandwidth available.

It is critical for an autonomous robot to complete its mission despite communication failure. The vehicle not only needs to know where it is supposed to go and what it is supposed to do but must also be able to determine its own safety through advanced obstacle avoidance and employ flexible "mission constraint" capability. A mission constraint is a limitation that must be considered and adhered to while planning and executing a subsea program. Constraints can be imposed by technical, operational, and environmental factors (such as weather or turbidity). Technical constraints are limitations related to the equipment or technology used—for example, battery limitations. Operational constraints are restrictions related to the procedures and protocols that must be followed. Minimum altitude restrictions, for example, could prohibit certain maneuvers from being performed due to vehicle safety concerns. Undesired events that can occur due to a vehicle fault or environmental conditions include communication loss, breach of minimum altitude, excessive pitch/roll, positioning accuracy, and sensor failure.

The ability to communicate via multiple media is essential, but this is somewhat complicated because the UID not only must be designed with multiple means of communication onboard but also be able to decide among them without rebooting or reconfiguring. The unit also must position itself correctly to communicate; so it must understand what behavior to execute to best enable each type of communication. For example, for ultra-short baseline (USBL) communication, the robot needs to be within the range of the receiver and at a depth that facilitates communication at a specific range.

Description and Application of Equipment and Processes

A development program was created to address the limitations of existing designs — improving maneuverability and introducing the ability to hover. The goal was to produce a UID capable of delivering rapid acquisition of high-quality data. The program, carried out over eight years, was supported at the outset by TotalEnergies and Chevron, ETC and later by Equinor (J. Cheramie and A. Anderson, 2023).

Initial development and trials, conducted 2014-2018, progressed software development and identified the limitations of adapting the existing vehicle design. In 2019 a new purpose-built solution was introduced, a hybrid vehicle with enhanced command and control capabilities to enable accurate, high-speed, single-pass, close-proximity inspections (A.S. Gower et al, 2022).

This project established a new methodology that expedites technology development, testing, validation, and approval by involving operators from inception through deployment to provide a high level of comfort for adopting serial-number-one technology. The goal was to design a unit that could meet these requirements and execute missions without human interaction while working subsea for extended periods (J. Cheramie and A. Anderson, 2023).

When the new, hybrid unit passed post-manufacturing acceptance testing, it underwent sea trials at Oceaneering's Living Lab autonomy center in Tau, Norway, in 2024.

At the Living Lab — a permanent, dedicated testing and development facility that was established in 2019 to advance subsea robotics and autonomous technologies — it was possible to develop, test, and qualify UIDs that are the next-generation autonomous underwater vehicles. With a permanent facility and the capability for daily testing and development, the methods of testing new theories can be controlled through established mission planning, simulation, risk assessment and mission execution processes. These processes enable engineers to predict parameters and constraints and modify settings before testing to ensure that "edge cases" are considered and mitigated. An edge case is a situation that occurs at the extreme end of the range of possible inputs, environmental conditions, or operating parameters. Edge cases often reveal bugs or weaknesses in systems that would not show up during normal use. In a critical system, handling edge cases is the difference between success and failure.

Essential Equipment

Typical UIDs combine the features of an ROV and an AUV, and in this case Oceaneering's Freedom was used. A built-in camera captures images in relatively clear environments, while a multibeam (acoustic) sensor allows the unit to "see" through murky water. The AUV/ROV also employs laser technology for rapid, accurate measurements. A bespoke vehicle framework within the software allows it to take inputs from all the sensors and decide which inputs should be used to direct flight control for the most accurate positioning (J. Cheramie and A. Anderson, 2023).

The hybrid vehicle operates in three modes. It can be piloted remotely via tether or through-water communication (to provide real-time control), or it can work autonomously, relying on designed-in software and hardware capabilities. In a typical UID mission, the unit follows a preprogrammed flight plan with rudimentary collision avoidance options. The UID is situationally aware of its environment and can react to unplanned events. Instead of following a preprogrammed path, it executes its mission by actively positioning itself and making decisions based on its surroundings and mission goals.

The design is uniquely flexible and can be optimized to meet diverse work scopes and requirements, including:

- High-speed, single-pass, close proximity autonomous pipeline inspection

- Autonomous free span and crossing inspection
- General visual and instrumented inspection of subsea structures (orbital)
- Bathymetric mapping using a multibeam echosounder
- Hi-resolution bathymetry mapping via laser
- Hi-resolution image mapping via a 12-megapixel camera.

A self-monitoring subsystem allows the UID to determine when to break away from a mission and return to the seabed docking station for system and mission defined parameters, The vehicle has a working range of up to 120 km, a working depth rating of 6,000 m, top speed of 1.75 m/s, and subsea deployment of up to six months.

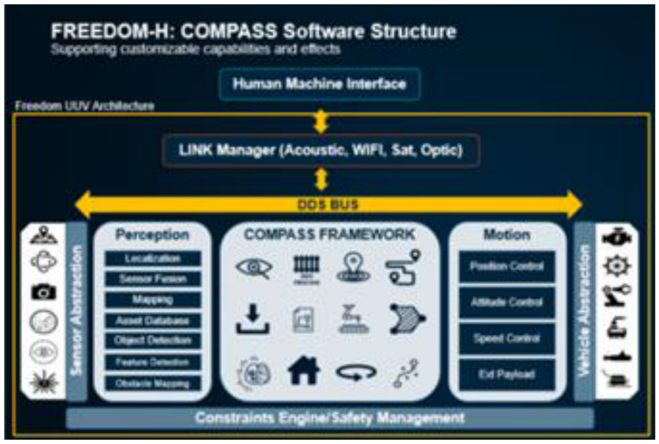
Test missions were designed and simulated using a proprietary Autonomy Simulation (Swimulator) system and then performed by the UID system in the Living Lab. Another proprietary technology, a mission review and visualization tool, allows engineers and operators to review, analyze, and visualize the actions and data collected during a robot's mission. Tool features include.

- Timeline-Based Playback with step-by-step mission replay (with pause, rewind, and fast forward), showing the robot's path, decisions, and sensor data.
- 3D Visualization of the robot's movement, showing seabed terrain and structures.
- Simplified navigation issue identification for positional errors and other anomalies.
- Event Markers for highlighting key events like obstacle detection, mission goal completion, or system faults for quick navigation to critical moments within the mission.
- Debugging and Diagnostics that allow log, command, and system state inspection at any point in the mission,
- Performance Metrics display for mission success, energy usage, localization accuracy, etc., supporting post-mission analysis and reporting.

The Freedom UID has long-range, advanced autonomy with payload-informed, feature-based navigation and the ability to stop, hover, orbit and reverse. Its performance relies on advanced, flexible communication systems.

Essential Software

The software for this unit — developed in a modular package that comprises reusable components and well-defined application program interfaces (APIs) — controls everything the vehicle does, including thrust and motion, hazard avoidance, data collection, and navigating. The software code was written in house and tested in a simulator, allowing developers to refine features and eliminate bugs before uploading it to the vehicle for in situ testing and validation.



The software is based on a modular microservices architecture bult on a robotics framework. An agile process was chosen to support rapid development, reusable libraries, and "pluggable" APIs for autonomy, mission behaviors, and most importantly, sensor inputs. The modularity of the perception stack of the framework allows rapid sensor integration, enables fusion algorithms for environment detection and navigation with situational awareness and permits multiple events to be triggered from one or multiple sensor streams.

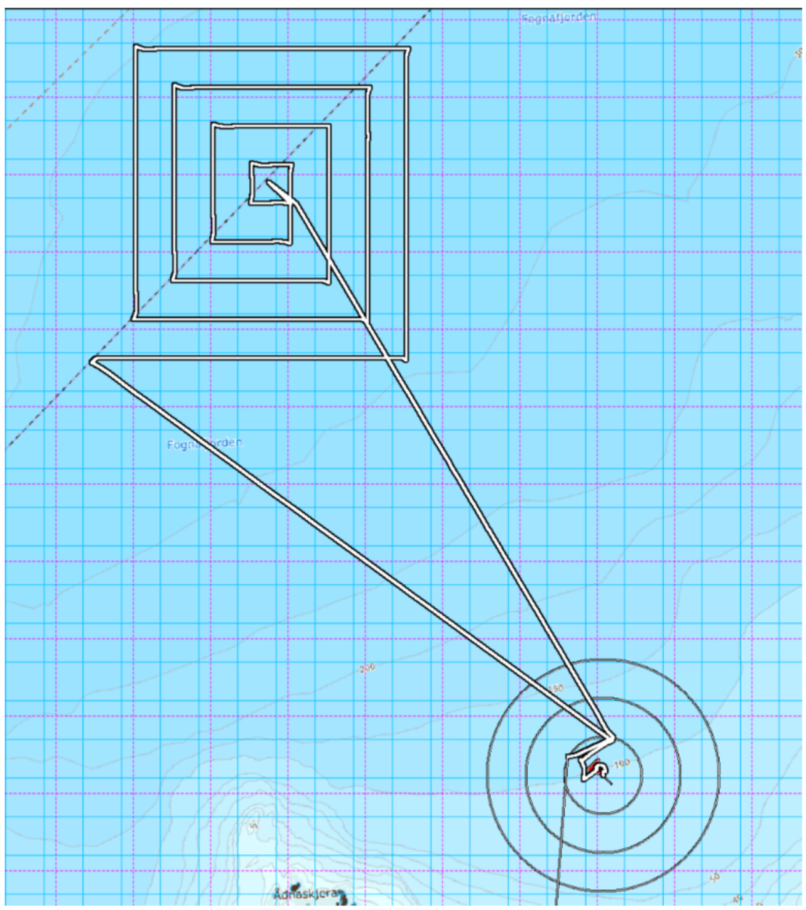
Data and Results

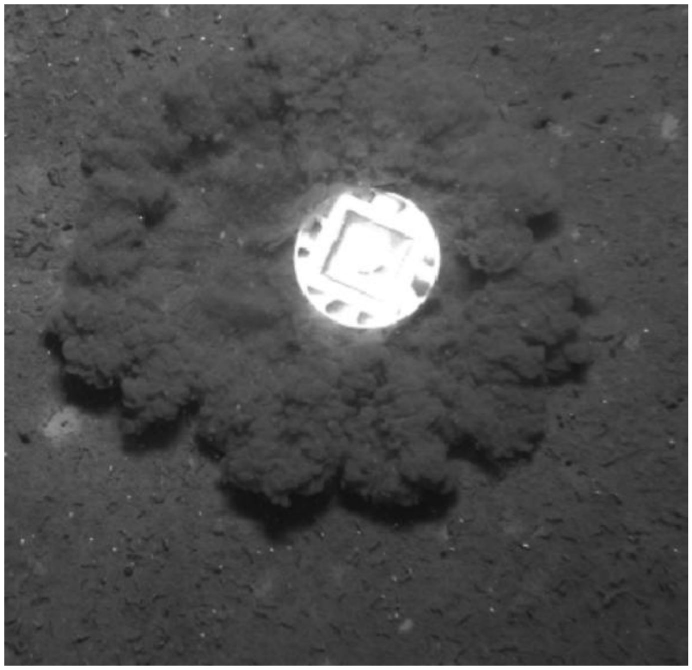
Task Element	Element Description and Outcome	Success / Fail
Mission Planning	All missions performed during the week were pre-planned and simulated.	Success
Initialization	A demonstration of the pre-dive checklist was given onboard the work vessel each day so operators and technicians could work through it and prepare the vehicle for operation.	Success
Launch	Equipment preparation, and Freedom AUV/ROV deployment within the cage. **Demonstrating Unique Vehicle Deployment Solution.**	Success
Undocking	With the cage at depth, the vehicle demonstrated its ability to undock from the cage and proceed to a hold point to await a release command to dive to depth from the operator. This hold was a practical measure to allow the cage to be recovered from the water. **Demonstrating Unique Vehicle Deployment Solution.** **Demonstrating Unique Flight Maneuverability Capability**	Success
Instruments Powered on	Once the vehicle reached the predetermined altitude, it powered on the Multibeam EchoSounder and Laser System. **Demonstrating efficiency measure.** **Demonstrating sequenced action without vehicle-to-operator communication.**	Success
The cage was removed from the water and the release command sent. From this point, the vehicle performed autonomously, without command or continuous communication with the surface.		

Demonstrating efficiency improvement measures implemented.		
Transit to Object Deployment Location	With the box pattern competed, the vehicle transited to a pre-defined location 400 m from the launch area at a speed of 1.5m/s, remaining in place upon arrival for a pre-defined, 1-minute period. **Demonstrating sequenced action without operator communication.**	Success
Hover Approach to Pre-determined Deployment Position	With the 1-minute hover-in-place achieved, the vehicle began a slow approach to the deployment position at an altitude of 20m. **Demonstrating sequenced action without operator communication.** **Demonstrating fine flight control.**	Success
Deployment of Small Payload	On arrival at the predetermined location, the vehicle triggered a magnet to release the payload. **Demonstrating sequenced action without operator communication.** **Demonstrating unique capability – payload deployment - automated as part of a mission instead of human triggered.**	Success
Controlled Hover to Departure Point and Transit to Pattern Survey Start	With the payload deployed, the vehicle moved slowly from the deployment location to a predefined departure point to avoid moving or disturbing the seabed around the object. **Demonstrating sequenced action without operator communication.** **Demonstrating unique capability – payload deployment - automated as part of a mission instead of human triggered.**	Success
Pattern Survey (Box Pattern)	The vehicle performed a pattern survey in the shape of progressively increasing boxes, showing control and line-following capability. Line length 4km, maximum speed 1.5m/s. **Demonstrating sequenced action without operator communication.**	Success
Transit to Recovery Position	On arrival at the end of the Pattern Survey, the vehicle transited at a higher speed (1.5m/s) to a pre-determined recovery location 400m away. **Demonstrating sequenced action without operator communication.** **Demonstrating unique capability – payload deployment - automated as part of a mission instead of human triggered.**	Success
Loiter in Place	When the unit reached its predefined end of line, it loitered in place at 65m depth, waiting for the operator to deploy the cage to 43m water depth.	Success
Recovery to Cage	The operator issued the release command to begin the docking process. The vehicle located the cage in the water column and orientated itself to the entrance but failed to enter on the first attempt. A second command re-initiated the docking mission, which was completed successfully. **Demonstrating Unique Capability – Recovery method.** **Demonstrating Unique Capability – Payload informed flight – Forward-looking camera.** **Demonstrating Unique Capability – Flight Maneuverability.** **Demonstrating Unique Capability – Dynamic interaction, fine control.**	Success

Charging and Data Offload from cage	Once the vehicle docked, the operator instructed a connector panel on the cage to move it into position to engage with connectors on the vehicle, with the process live streamed from the vehicle's forward-looking camera. **Demonstrating Unique Capability – Recharging, subsea data exchange.**	Success
GPS Position Update Behavior	The onboard GPS antenna fed an updated position to the surface INS. (Not Demonstrated - Vehicle surfacing/diving after GPS position update).	Success

Missions that do not require communication between human operator and vehicle <6 hours in duration have been demonstrated and validated, but this requires detailed mission rehearsal. The next challenge is for the vehicle to provide data bursts with mission details to the remote operations center so mission progress can be documented even when no additional instruction is required.

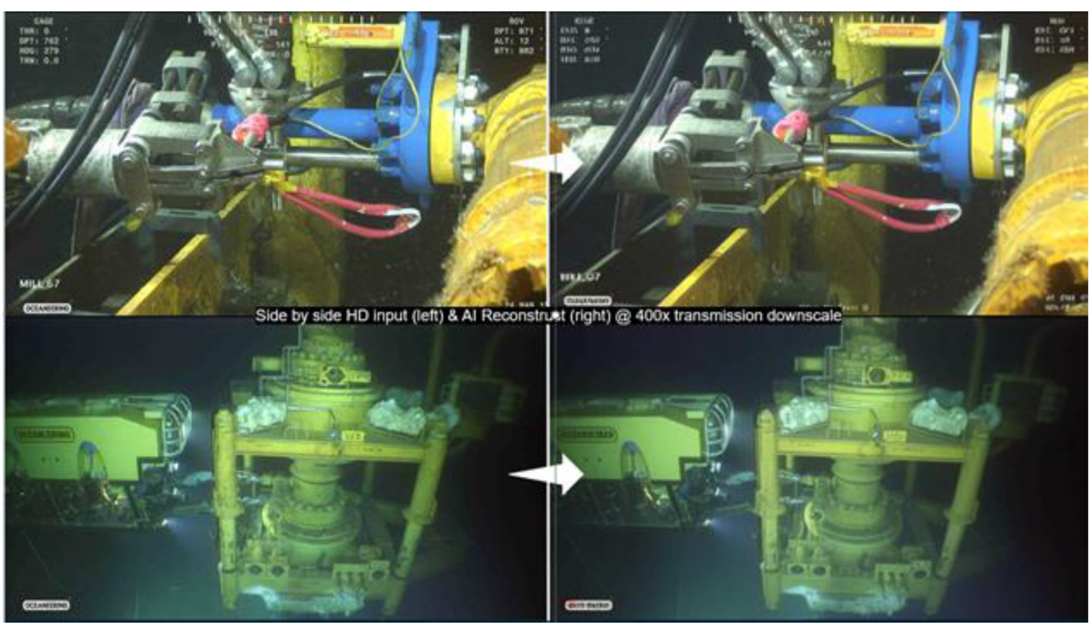




This photograph shows the path followed for the mission executed as well as the object dropped during the mission.

Lossy file compression with AI-based image reconstruction

Long-range communication remains a challenge. Compressing files to very small sizes to expedite file transmission is the next development step. Experts are looking for ways to apply ML to improve visual quality of decompressed images even in cases where significant data has been discarded as part of the file compression process. The technique — commonly used for images and video, where perfect accuracy is not always necessary for human perception — is Lossy File Compression, which reduces the file size by permanently removing some of its data. Oceaneering has been able to compress images by up to 400 times using this technique, making files much easier to send via extremely low bandwidth.



Precisely reconstructed images recorded by the UID following compression and low-bandwidth transmission provide clear detail of the underwater environment.

A 6-degree-of-freedom UID has the unique ability to stop, hover, orbit and reverse, it can capture images of objects and structures that can be converted to digital still images — complete with positional and heading reference data — that can be transmitted using a very low bandwidth communication link.

Clear images can be created through the application of Gaussian splatting, a modeling technique for 3D scene reconstruction and rendering that uses individual still images to create highly realistic, interactive models. This technique uses large, single images consisting of millions of geospacial dots and applies AI to remove 99% of the point cloud (without losing any subjective model information) to produce a smaller file. Photographs taken at fixed intervals around a structure are used to compute the 3D model on the vehicle. On the receiving end, images are reconstructed using AI models. This approach allows anomalies to be identified by the system during the survey and communicated to a remote operations center even a great distance away with very low baud rate within 5-10 minutes instead of the 1-4 hr required without compression.

Applying advanced communication technology allows the file size to be reduced and the images reconstituted using AI to produce detailed images that remotely housed experts can used to visually assess the vehicle mission or asset status in real time.

Conclusion

UIDs are still evolving, especially in terms of autonomy, reliability, and communication, but much work has now been carried out to demonstrate the technical readiness of the technology for pilot projects.

Expanding the capability of ROVs to UIDs that can operate independently with only human supervision will deliver benefits, including:

- Reduced Cost (less reliance on large, crewed vessels).
- Increased Safety (minimizes human exposure to dangerous environments).
- Operational Efficiency (long-term, continuous and autonomous operation).
- Rapid Deployment (launched from smaller vessel or by unmanned surface vehicles).

References

OTC-31733-MS, Subsea Robotics - AUV Pipeline Inspection Development Project (Gower Stephen Andy and Lardeux Mathieu, TotalEnergies; Gilmour Bill and Ling Damian, Chevron ETC).

OTC-32559-MS, Developing the Next Generation of Pipeline Inspection AUV (Cheramie Jami and Anderson Alan, Oceaneering International)